EdaphNet 2.0

COMPLETE PROJECT DOSSIER · VERSION 2.0

Objectives · Methodology · Datasets · Models · Prior work · Gaps · Contributions
Experiments · Evaluation metrics · Results · Architecture · Development plan

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Every entry above is a clickable link in the PDF. Every number in this dossier comes from the project's committed documents (docs 08–26), the paper draft v0.2 and the run records in `results/runs/`.

1. The project at a glance

EdaphNet is a research project, with working software, that asks whether a cheap, independent observation of a soil sample (mainly its **colour**, as a phone camera could record it, or a laboratory **spectrum**) can tell a soil-testing laboratory **which reports to re-test** and **what went wrong** with them. It checks a whole laboratory batch at once for two kinds of mistakes: a report that belongs to another sample (a **swap**) and a single corrupted number (a **value error**). This dossier is the complete record of the project: what it is for, how it was built, what data and models it uses, what was tested and measured, what is new, what failed, and what happens next.

Table 1. Project status at the date of this dossier.

Item	Status
Research	Complete as a first study: simulation, two real datasets, a pre-registered test on new data, exploratory analyses, review fixes.
Main pre-registered result	With a colour witness the method did **not** beat the best existing method (+0.014 [−0.008, 0.038]); with a spectrometer it did (+0.092 [0.069, 0.113]). Attribution passed: correct error type 97–100% vs 24–29%.
Scale of the evaluation	23,219 real soils (KSSL 19,746; ICRAF-ISRIC 3,473) · 5 witness levels · 8 comparison methods · 50 settings in the main test · 5 seeds · about 7,200 reports per setting
Paper	Draft v0.2 with 6 figures; framing "When is a cheap witness enough?"; target: mid-tier applied journal.
Code	≈ 1,520 lines of method code (14 modules in `src/edaphnet2`, including combined policy), 29 scripts, ≈ 800 lines of app code, 8 test suites (≈ 560 lines of tests), demo and report pages.
Verification	39 automatic tests passing across 8 suites (plus a slow resume-equivalence test); dataset rebuild from the download verified identical; one seed reproduced exactly.
Records	21 numbered documents, a decision log with 25 dated decisions, 59 git commits, a run record for every experiment.
Demonstrators	Animated live demo (photo + report → verdict, guided mode), offline presentation page, command-line demo, separate farmer app (trained leaf disease prototype head across 28 classes, crops, soil colour, report check).
Data status	Real soils and real laboratory values; **planted** errors; colour **calculated** from laboratory spectra (no real phone photos yet; no Indian soils yet).
Readiness	Research prototype, validated in simulation and on public datasets only. Not a laboratory product.

How to read this dossier. Sections 2–4 set out the problem, the objectives and the scope. Section 5 reviews existing work and Section 6 states the gaps that follow from it. Section 7 lists the contributions claimed and what is not claimed. Sections 8–11 describe the architecture, the data, the method and the models. Sections 12–14 describe the experiments, the metrics and the results, including the failures. Sections 15–17 give the limitations, the integrity record and the intellectual-property position. Section 18 is the development plan and Section 19 explains how to reproduce every number.

2. Problem statement and motivation

2.1 Soil reports drive real decisions

A soil test report decides how much fertiliser and lime a farmer applies and, increasingly, how much carbon a field is credited with. Public programmes test soils at very large scale. Laboratories control quality with reference samples, proficiency testing and plausibility rules, but these work at the level of the laboratory. The individual farmer who receives one report has no independent way to know whether that report is right for *their* sample. Measurement disagreement is not rare: in one comparison of eight service laboratories, 28% of soil organic carbon results fell outside the acceptable range of the North American Proficiency Testing programme (cited in the paper draft; to be verified against the full text).

2.2 Failure 1: swapped samples are invisible to every lab-only check

Samples are mislabelled, placed in the wrong tray position, or a whole tray is shifted by one place. The farmer then receives the results of another soil. A swapped report looks perfectly normal, because it *is* a real report of a different soil: every value is plausible, every internal rule is satisfied. Only something that observes the physical sample itself can reveal the mismatch.

2.3 Failure 2: corrupted values

Single numbers are corrupted by decimal slips (×10, ÷10), unit confusion, transposed digits (0.16 → 0.61), interchanged columns, or analytical failures (gross errors). Some of these are caught by range checks; many are not, because the wrong value can still be plausible.

2.4 Failure 3: re-testing everything is unaffordable

Re-analysing every sample doubles the cost of testing. A laboratory can afford to re-test only a small share of reports, so it needs a **ranked list**: which few reports are most likely wrong, and what to look for when re-testing them.

2.5 Why this is a research problem and not only an engineering one

A cheap witness such as soil colour is **weak**: it explains carbon well but most other properties poorly. Whether such a weak signal is enough to find errors, which errors it can find, how it should be combined across a batch where swaps involve several reports at once, and whether false alarms can be controlled with realistic calibration data are open questions. The answers are not obvious in advance, and the project shows that some intuitive answers are wrong (for example, that the joint model's advantage would be largest for the weakest witness).

3. Research objectives and questions

3.1 Aim

To design, build and evaluate, with rules fixed in advance, a method that uses an independent observation of each soil sample to rank the reports of a laboratory batch for re-testing, explain what went wrong, and propose a correction, and to establish **when such a cheap witness is enough**.

3.2 Objectives

Table 2. Research objectives and their current status.

ID	Objective	Status
RO1	Search existing papers, patents and products; define what would be new.	Done (informal search); formal Scopus/WoS search pending
RO2	Specify the problem formally: error model, swap patterns, baselines, hypotheses, pass/fail rules.	Done (doc 11, pre-registered)
RO3	Build a joint batch model over swaps and typed value errors driven by a witness model.	Done (`src/edaphnet2`)
RO4	Test the mechanism on synthetic data with controllable witness strength.	Done (doc 13)
RO5	Measure how much colour and spectra reveal about real soil properties.	Done (docs 14, 21)
RO6	Test the method against strong baselines on real soils, with pre-registered rules.	Done; main claim failed with colour, passed with spectrometer
RO7	Study false-alarm control with realistic calibration data.	Done; calibration mistake found and fixed
RO8	Map which errors each witness can make detectable.	Done descriptively (docs 19, 26)
RO9	Make the work reproducible and demonstrable.	Done (run records, REPRODUCE.md, demos)
RO10	Validate with real phone photos and real laboratory errors (Indian soils).	**Not started**: the principal outstanding objective (paper 2)

3.3 Research questions

- **RQ1.** Can an independent observation of the sample make report errors detectable that lab-only checks miss, and which ones?
- **RQ2.** Does a joint batch model over swaps and value errors find more errors than the best existing method within a fixed re-test budget, and for which witness strength?
- **RQ3.** Can the model say correctly *what* went wrong (swap partner, error type)?
- **RQ4.** Can false alarms be controlled at a stated rate with realistic calibration data, and what does that require?
- **RQ5.** How does the answer to RQ2 depend on the mix of swaps and value errors, and on error types the model does not know?

3.4 How the question changed (all changes recorded)

Table 3. Versions of the main claim and their outcomes.

Version	Claim	Outcome
v1 (doc 11)	A phone-grade colour witness catches more errors at a guaranteed 10% false-alarm rate.	Failed (doc 17), partly due to our calibration mistake
v2 (doc 18, pre-registered)	With a 5% re-test budget, the method finds more errors than the best existing method, using colour.	Failed on new data (KSSL, doc 21)
Current framing	When is a cheap witness enough? Detectability, attribution, error mix, calibration requirements.	Supported by the evidence

4. Scope and boundaries

4.1 In scope

- **Errors:** sample swaps (neighbour swaps, tray shifts, random pairs) and single-value errors (×10, ÷10, unit confusion, digit transposition, column interchange, gross analytical error); in a separate test, four error types the model does not know.
- **Witnesses:** none (W0); soil colour in CIELAB with phone-like noise ΔE 0, 2, 4, 8 (W1); 10 visible bands (W2); the full Vis-NIR spectrum (W3).
- **Data:** open soil spectral libraries with real laboratory values (KSSL: organic carbon, nitrogen, total carbon; ICRAF-ISRIC: eight properties).
- **Output:** a suspicion score, an explanation and a proposed correction for every report; a re-test list at a fixed budget; optionally flags with a target false-alarm rate.

4.2 Deliberately out of scope in this study

Table 4. Exclusions and the reason for each.

Excluded	Reason
Real phone photographs	No public dataset pairs phone photos with laboratory reports of the same samples; planned for paper 2 (doc 15).
Real laboratory errors	Unknowable without blind duplicates or laboratory re-test records; planned for paper 2.
Indian soils	No open Indian soil spectral library with the needed properties; planned pilot with an Indian laboratory.
LUCAS (European) data	Needs an access request; planned as a confirmation study (doc 20).
Changes to the method after the pre-registered test	The method was frozen (commit 095deaa) so the test could not be tuned afterwards.
The farmer app (leaf disease, crops)	A separate application; not part of the research claims (Section 8).

4.3 What the project explicitly does not claim

It does not claim that a phone photo can replace a laboratory test, that colour alone beats existing checks, that the method has been validated on real photographs or real laboratory errors, or that any number applies to Indian soils. Every result is labelled as pre-registered or exploratory.

5. Existing work

5.1 Search method

A targeted search of papers (Google Scholar, publisher sites, arXiv), patents (Google Patents) and vendor material was run in step 1 (docs 09, 10). Sixteen prior works were compared against the nine elements of the candidate method. A formal, logged search in Scopus or Web of Science has **not** yet been run and is required before submission; the "no equivalent found" statements below are therefore provisional.

5.2 Prior-art matrix

Table 5. What prior work covers. E1 flag a lab value via a prediction · E2 independent sensing of the same specimen · E3 cheap/weak sensor · E4 error-type attribution · E5 corrected value · E6 swap detection · E7 joint swap + value-error inference over a batch · E8 false-alarm guarantee · E9 soil domain. ● covers · ◐ partly · ○ no · ? unverified.

#	Work	E1	E2	E3	E4	E5	E6	E7	E8	E9
P1	US 8,352,394 B2 (lab validation via predicted values)	●	◐	○	○	○	○	○	○	○
P2	Zhou et al. 2023, errors in a numerical response (cleanlab)	●	○	○	○	○	○	○	○	○
P3	Bates et al. 2023, conformal outlier testing	◐	○	○	○	○	○	○	●	○
P4	Jin & Candès 2023, selection with conformal p-values	◐	○	○	○	○	○	○	●	○
P5	Peng et al. 2025, conformal prediction with cellwise outliers	◐	○	○	○	◐	○	○	◐	○
P6	Clinical wrong-blood-in-tube ML (AJCP 2018; Clin Chem 2023)	●	○	○	○	○	●	○	○	○
P7	US 7,783,582 (Bayesian clinical-lab error detection)	●	○	○	◐	?	?	○	○	○
P8	Clinical unit harmonisation framework (2025)	◐	○	○	●	●	○	○	○	○
P9	Broman et al. 2014, sample mix-ups in genomics	●	●	○	○	○	●	◐	○	○
P10	Crosscheck (ENCODE) multi-omics matching	●	●	○	○	○	●	○	○	○
P11	MIR spectra flag wet-lab inconsistencies (KSSL)	●	●	○	?	?	?	○	○	●
P12	Schmidinger et al. 2026, uncertainty-based rejection (Vis-NIR)	◐	●	○	○	○	○	○	○	●
P13	LUCAS: photo of each sample for manual checking	◐	●	●	○	○	◐	○	○	●
P14	Soil laboratory information systems (rules, barcodes)	◐	○	○	○	○	◐	○	○	●
P15	Smartphone soil-property papers; LandPKS	○	●	●	○	○	○	○	○	●
P16	CN108732129B (soil components from images)	◐	●	?	○	○	○	○	○	●

A recent open dataset evaluates smartphone imaging for soil organic carbon across four phones on 84 samples (Zenodo 10.5281/zenodo.22876123, 2026). It predicts properties; it does not check reports. No soil laboratory-error benchmark was found (LimeSoDa and REIN are the nearest; neither contains laboratory errors).

5.3 What the field looks like as a whole

Three traditions exist and do not meet. **Proximal soil sensing** (smartphone colour, spectroscopy) predicts properties but never asks whether a laboratory report is right. **Laboratory and clinical quality control** checks values against predictions or history, and genomics detects sample swaps, but with near-perfect witnesses or with patient history rather than an independent look at the same specimen. **Statistical error detection** (cleanlab, conformal outlier testing) provides general tools without the structure of laboratory errors (trays, typed slips). EdaphNet sits where the three meet: a weak, independent observation of the specimen, used jointly across a batch, with typed errors and a stated false-alarm rate.

6. Identified research gaps

Each gap is stated as an absence in the work reviewed in Section 5, with the closest existing work. Gaps marked **addressed** are answered by this project; the remainder are open and appear in the development plan.

Table 6. Research gaps, closest prior work and status.

#	Gap	Closest existing work	Status here
G1	No method uses an independent observation of the same soil sample to check laboratory reports automatically.	LUCAS manual photo check (P13)	Addressed (C1)
G2	No method detects swaps and value errors jointly across a batch.	Broman 2014 (swaps only, strong witness)	Addressed (C1)
G3	Detection methods do not say what went wrong or propose a correction.	Unit harmonisation (P8), units only	Addressed (C2): 97–100% correct type
G4	No map of which report errors a witness of given strength can reveal.	None found	Addressed descriptively (C3)
G5	No guidance on when a cheap witness pays off relative to the error mix.	None found	Addressed exploratorily (C3); needs confirmation
G6	Conformal false-alarm control not studied for report verification with realistic calibration data.	Bates 2023 (general)	Addressed (C4)
G7	Evaluations of soil checks are not pre-registered and rarely report failures.	Most literature	Addressed (C5)
G8	No benchmark of soil laboratory-report errors.	LimeSoDa, REIN (no lab errors)	Partly: planted-error benchmark built; real errors open
G9	No evaluation with real phone photos paired with laboratory reports.	Zenodo 2026 (84 samples, prediction only)	Open (paper 2)
G10	No evaluation on Indian soils or Indian laboratory practice.	None found	Open (paper 2)

7. Novel contributions

The contribution claimed is a combination plus a set of findings. Several components have analogues in the literature (Section 5); the combination, and the evidence about when it works, does not appear in any work retrieved so far.

Table 7. Claimed contributions, the gaps they close and where the evidence sits.

ID	Contribution	Gaps	Evidence
C1	**Joint batch verification with an independent witness.** Every report in a batch is scored under structured swap hypotheses and a library of typed value errors, using a calibrated model of what each sample's witness implies.	G1, G2	§14.5–14.8; swaps found +0.10 over the best baseline with colour, +0.23 with a spectrometer
C2	**Explanation and correction.** For each flagged report, the most likely swap partner or error type and a proposed corrected value.	G3	§14.13; 97–100% correct error type (pre-registered A3) vs 24–29% for rule-based repair
C3	**Detectability and error-mix map.** Which errors each witness strength reveals, and the share of swaps at which a cheap witness pays off.	G4, G5	§14.8, §14.10; colour pays off from 60–70% swaps, a spectrometer from 30%
C4	**Calibration finding.** Reusing calibration samples silently breaks conformal false-alarm control; power depends on distinct calibration reports relative to the number tested at once.	G6	§14.4, §14.11; realised FDR 0.57 → 0.00 after the fix; the m/(q(N+1)) bound
C5	**Pre-registered, reproducible evaluation with negative results.** Rules committed before tests; run records for every number; failures reported.	G7, G8	§16, §19; docs 11, 18, 21

Statement of what is not claimed. Flagging values via predictions, conformal p-values with Benjamini–Hochberg, cross-modal swap matching, unit-error tables and phone soil colour are existing tools used as components. The method does **not** beat the best existing method with a colour witness at an equal mix of errors (pre-registered A1 failed). The structured swap prior gave no benefit (H6 not supported). The exploratory results (error mix, per-batch control, gross-error diagnosis) are hypotheses for confirmation, not proven claims. A reviewer may call C1 "Broman + Bates + a unit table"; the defence is C2 and C3, which no component alone provides.

8. System architecture

8.1 Overview

The project has one research pipeline, which produces every number in the paper, and three side branches that reuse parts of it: the demonstrators, the preparation for paper 2 (real photos), and a separate farmer application. The method itself sits in one package, `src/edaphnet2`, frozen before the pre-registered test.

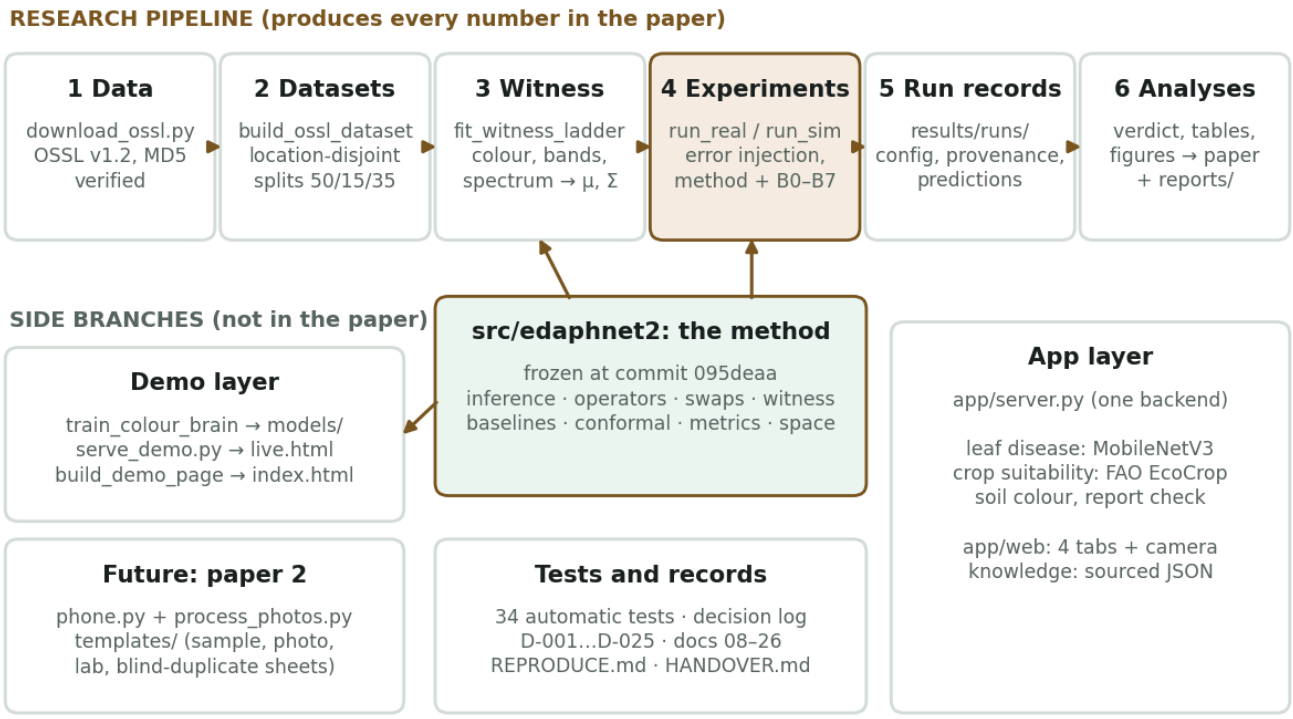


Figure 1. System architecture: the research pipeline (top) and the side branches (bottom).

Table 8. Components, their location and their responsibility.

ID	Component	Where	Responsibility
D1	Raw data	`data/raw/ossl/`	Three OSSL v1.2 files (soil sites, laboratory values, Vis-NIR spectra), MD5-verified.
D2	Datasets	`data/processed/`, `data/splits/`	Clean datasets with location-disjoint train / calibration / test splits; witness predictions.
P1	Data build	`download_ossl.py`, `ossl_audit.py`, `build_ossl_dataset.py`	Download, audit, clean, split.
P2	Witness ladder	`fit_witness_ladder.py`	Colour from spectra, phone-like noise, witness models, R^2 on unseen locations.
M1	Inference	`src/edaphnet2/inference.py`	Joint batch scoring, explanation, correction.
M2	Error library and swaps	`operators.py`, `swaps.py`	Typed value errors and swap patterns; error injection.
M3	Witness models	`witness.py`	Predicted property distribution per sample.
M4	Baselines	`baselines.py`	B0–B7 comparison methods and B6 rule-based repair.
M5	Error control and metrics	`conformal.py`, `metrics.py`	Conformal p-values, Benjamini–Hochberg, recall at budget, FDR, per pool.
E1	Experiment runners	`run_sim.py`, `run_real.py` + `configs/`	Settings × seeds; memory-lean; resumable with identical results.
R1	Run records	`results/runs/`, `tracking.py`	Config, code version, environment, metrics, every report's scores.
A1	Analyses and figures	`analyse_amendment.py` and 8 analysis scripts	Verdict, exploratory tables, figures; outputs to `results/exploratory/`.
U1	Demonstrators	`serve_demo.py`, `demo_site/`	Live demo and presentation page.
U2	Farmer app	`app/`	Separate application; not in the research claims.

8.2 Experiment flow

One experiment setting runs as follows: the runner loads the dataset and witness predictions for the chosen witness rung → splits test samples into batches of 24 in random or clustered order → plants swaps and value errors with a seeded random generator → scores every batch with EdaphNet, its four ablations and the seven baselines → computes conformal p-values from clean calibration reports → measures recall at a 5% budget and power/FDR at q = 0.10 per pool of 50 batches → writes all scores and metrics to the run folder → moves to the next setting. Seeds are derived from (seed, rung, ordering) with a stable hash, so an interrupted run resumes with identical numbers.

8.3 Technology stack

Table 9. Technology choices and the reason for each.

Layer	Technology	Reason
Language	Python 3.14	Scientific ecosystem; one language for method, experiments and demos.
Numerics	NumPy 2.5, SciPy 1.18, pandas 3.0	Exact enumeration of hypotheses; chunked reading of large prediction files.
Machine learning	scikit-learn 1.9, LightGBM 4.7, cleanlab-style regression scores	Small, fast, CPU-only models; strong tabular baselines.
Colour science	colour-science (CIELAB from spectra, D65, 2°)	Standard colorimetry; ΔE noise model for phones.
Figures and PDFs	matplotlib; headless Chrome for printing	Every figure drawn from run records; no PDF library needed.
Demo server	Python standard-library HTTP server; plain HTML/CSS/JS	No framework to install; runs offline on the laptop.
App (separate)	PyTorch / torchvision (MobileNetV3), Pillow	Leaf-disease features on CPU.
Testing	pytest (39 tests + 1 slow resume test)	Method, data, photo processing, app, combined re-test policy.
Hardware	Laptop, CPU only, 7.8 GB RAM	Drove the memory-lean, resumable runner design.
Version control	git + GitHub (private)	Pre-registration by commit; provenance in every run.

8.4 Repository layout

Table 10. Top-level folders.

Folder	Contents
`src/edaphnet2/`	The method, baselines, combined policy, error control, metrics, data handling, run tracking, photo processing (14 modules).
`scripts/`	Data build, experiments, analyses, figures, PDFs, demos, app helpers (29 scripts).
`models/`	Trained model artifacts (`leaf_head.joblib` prototype across 28 classes, `colour_brain.joblib`).
`configs/`	Experiment settings (5 YAML files).
`results/`	Run records, exploratory tables, data audits, logs.
`docs/`	Documents 08–26, decision log, search log, paper draft and figures.
`data/`	Checksums and splits (tracked); raw and processed data (local only, rebuildable).
`reports/`	PDFs: paper, progress report, faculty discussion, summaries, this dossier.
`demo_site/` , `app/` , `templates/` , `tests/`	Demonstrators; farmer app; step-6 collection sheets; automatic tests.

9. Datasets

9.1 Sources used

Table 11. Datasets in use, with size, provenance and licence.

Dataset	Size	Properties	Used for	Licence
KSSL Vis-NIR (Open Soil Spectral Library v1.2)	19,746 samples, 6,999 locations	Organic carbon, total nitrogen, total carbon; Vis-NIR spectrum	**Pre-registered test**; unknown-error test; live demo	CC-BY
ICRAF-ISRIC (OSSL v1.2)	3,473 samples, 553 profiles	pH, OC, clay, sand, CEC, K, Ca, Mg; Vis-NIR spectrum	First real test; calibration diagnostic	CC-BY
Synthetic T0	Generated (30 settings)	8 properties (Gaussian copula) with a colour-like witness of chosen strength λ	Mechanism test	—

9.2 Sources wanted but not obtained

Table 12. Datasets identified as desirable but not used.

Dataset	Why it was not used	What it would add
LUCAS topsoil (EU)	Needs an access request (free); not yet submitted	A second large dataset (≈ 19,000 samples, 2018) for a confirmation test
Phone photos paired with lab reports	None public with reports of the same samples	A real-photo test; the Zenodo 84-sample smartphone set (CC BY 4.0) is a candidate, not yet downloaded
Real laboratory errors	Not knowable without blind duplicates or re-test records	Real error rates and mix (paper 2)
Indian soils (e.g. Soil Health Card laboratories, VIT VAIAL)	No open Indian soil spectral library; collaboration needed	Transfer to the target setting
Proficiency-testing data (WEPAL, NAPT)	Access not checked	Real inter-laboratory deviations

9.3 Data pipeline and quality controls

Table 13. Pipeline steps and the failure each one prevents.

Step	What it does	What it prevents
Download	Fetches the three OSSL files and checks MD5 checksums against the tracked manifest.	A changed or truncated source file (checksums verified against the online bucket on 3 October 2026).
Audit	Counts usable samples per dataset and property; checks licences.	Using rows without spectra: an early count included 23,852 empty spectrum rows; corrected (doc 12 §7).
Build	Selects the property set, harmonises units, removes incomplete rows.	Mixed units and missing values entering the model.
Split	Assigns train 50% / calibration 15% / test 35% by profile or location, with a fixed seed.	Optimistic results from testing on soils from places seen in training.
Witness	Computes CIELAB colour from each spectrum; adds phone-like noise; fits witness models with out-of-fold residuals.	Witness models that have seen the test samples.
Calibration	Uses one score per distinct calibration sample.	Pseudo-replication that inflated false alarms (Section 14.4).
Verification	Rebuild of KSSL from the downloaded files compared with the stored dataset.	Silent drift: all 10 arrays and 19,746 samples identical.

9.4 What is real and what is simulated

Table 14. Real versus simulated components.

Component	Status	Details
Soil samples, laboratory values	Real	Measured in real laboratories (KSSL, ICRAF-ISRIC).
Spectra	Real	Each sample's Vis-NIR spectrum from a laboratory instrument.
Errors in reports	Simulated	Planted by the code (swap event in 20% of batches; value errors in 2.5% of other reports; ≈ 5% erroneous reports), so the truth is known. Standard practice for measuring detection.
"Phone" colour	Simulated	Calculated from each laboratory spectrum, plus camera-like noise ΔE 0–8. A best case: dried, ground soil in perfect light.
Synthetic soils (T0)	Simulated	Only for the mechanism test; main results do not depend on them.

9.5 What the data look like

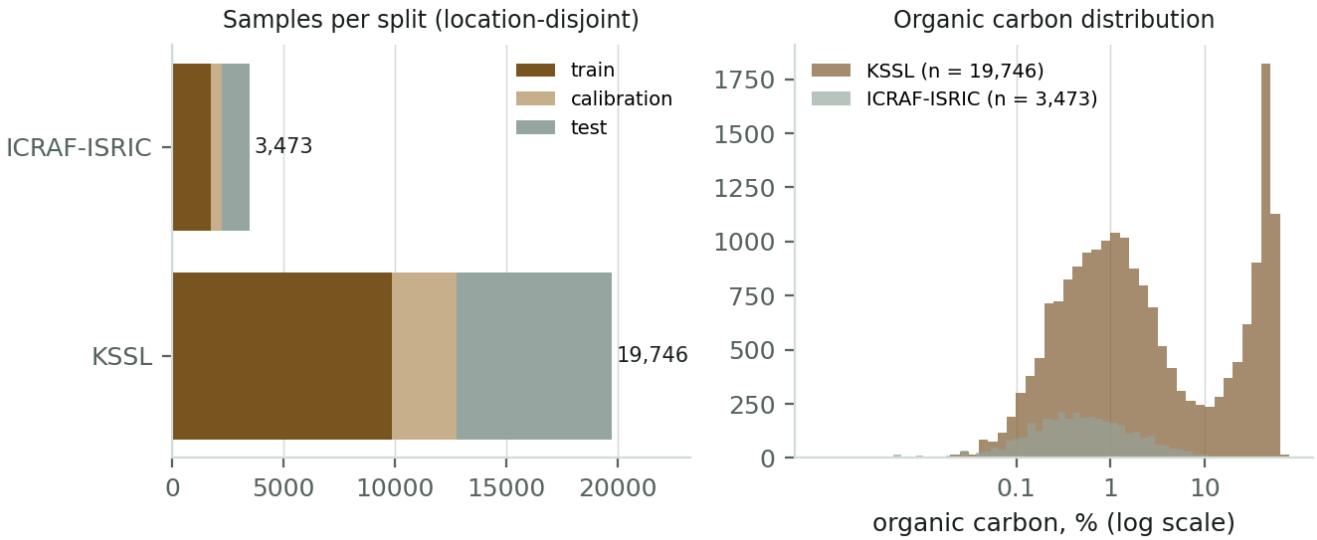


Figure 2. Left: samples per split. Right: organic carbon on a log scale; the KSSL peak above 20% comes from organic (peat) horizons.

KSSL covers a wide range of organic carbon, from mineral subsoils (≈ 0.1%) to organic horizons (> 30%), which is why colour, which mostly tracks darkness and therefore carbon, explains KSSL carbon well. ICRAF-ISRIC is smaller but covers more properties that colour does not see (pH, CEC, potassium, texture).

10. Methodology

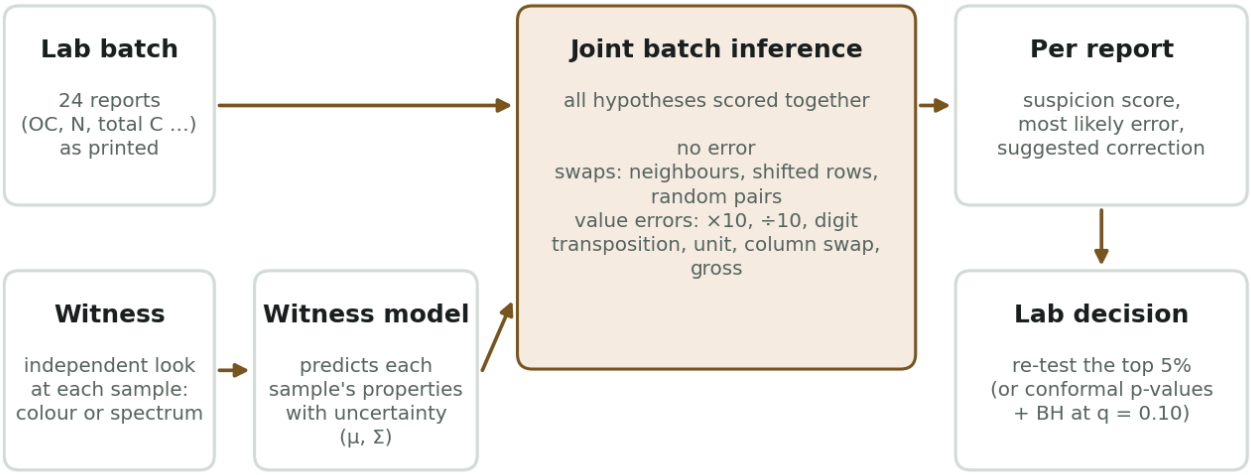


Figure 3. The method in one picture.

10.1 Witness model

Each property x is mapped to a standardised latent scale z (identity, log or logit, per property). Given the witness w of a sample (its CIELAB colour, 10 visible bands, or 30 principal components of the spectrum), the predicted latent properties are

$$p(z \mid w) = Normal(\mu(w), \Sigma)$$

where μ(w) is a gradient-boosted regression per property and Σ is the full covariance of out-of-fold residuals. Because Σ is full, the model knows how carbon, nitrogen and total carbon move together, so one corrupted value can show up as an inconsistency with the others even where the witness says little about that property.

10.2 Value-error library

Table 15. The value-error library $\mathcal{C}$ (at most one corrupted value per report; equal prior per group, uniform within a group).

Error group	Operation on the printed value	Example
Decimal slip	× 10 or ÷ 10	OC 0.82 → 8.2
Unit confusion	Conversion between units used for that property (e.g. % vs g/kg)	OC 1.2 % printed as 12 (g/kg)
Digit transposition	Two adjacent digits of the printed number exchanged	0.16 → 0.61
Column interchange	Values of two properties exchanged	N and OC columns swapped
Gross analytical error	Multiplicative factor on a grid in [1/3, 0.8] ∪ [1.25, 3]	OC measured 2× too high

10.3 Swap hypotheses

The enumerated set: identity (no swap); adjacent transpositions; cyclic rotations by ±1 (blocks of length ≥ 3) and ±2 (length ≥ 5), which model a tray shifted by one or two places; and arbitrary pair transpositions. A structured prior gives tray-like patterns more weight; a uniform prior is tested as an ablation.

10.4 Joint batch inference

For report j and candidate sample i, the evidence that report j was measured on sample i, allowing any corruption c, is

$$\ell(j, i) = \log \Sigma_c P(c) \cdot p(T_c^{-1}(r_j) \mid w_i) \cdot |J_c|$$

where T_c is the corruption, r_j the printed report and $|J_c|$ the Jacobian of the transformation. The log-posterior of a swap hypothesis π is

$$\log P(\pi) + \Sigma_{j \text{ moved by } \pi} [\ell(j, \pi(j)) - \ell(j, j)]$$

normalised over the enumerated set, so inference is exact over that set. For each report: P(clean) = (1 − P(moved)) · P(c = none | r_j, w_j); the **suspicion score** is 1 − P(clean); the **explanation** is the most likely swap partner or the most likely corruption; the **correction** is the inverse of that corruption.

10.5 Re-test policy

Reports are ranked by suspicion within a pool of 50 batches (about one laboratory day), and the top 5% are re-tested (the primary metric of amendment v2). Ties are broken at random with a fixed seed.

10.6 False-alarm control

For a stated false-discovery rate, each test report gets a conformal p-value against N clean calibration reports, p = (1 + #{calibration scores ≥ s}) / (N + 1), and Benjamini–Hochberg is applied at q = 0.10 per pool. Each distinct calibration sample contributes **one** score (Section 14.4). Weighted and per-group (Mondrian) calibration were also tested for shift between batches.

10.7 Error injection protocol

In each test batch of 24: with probability 0.2 a swap event (adjacent swap, tray shift or pair swap, equally likely); every other report receives a value error with probability 0.025, drawn from the library (or, in the unknown-error test, from four types outside it: ×100, gross ×4–8, a value copied from the neighbouring report, a dropped digit). About 5% of reports are erroneous, roughly half of them swaps. Batches are formed either in random order or clustered (layers of the same profile or location together), the harder case.

11. Models

11.1 Witness models

Table 16. Witness model specification.

Property	Value
Task	Regression of each latent property on the witness; Gaussian predictive distribution with full residual covariance
Inputs (ladder)	W0 none · W1 CIELAB colour with noise ΔE 0 / 2 / 4 / 8 · W2 10 visible bands · W3 30 principal components of log(1/R)
Learner	LightGBM: 300 trees, learning rate 0.05, 15 leaves, minimum 20 samples per leaf, subsample 0.8, column sample 0.8
Training data	Training split only; residual covariance from out-of-fold predictions
Reported quality	R^2 on test locations; Gaussian and conformal 90% coverage

11.2 EdaphNet and its ablations

Table 17. The method and the ablated variants used to measure what each part contributes.

Variant	What is removed
Full (EdaphNet)	Nothing: swap hypotheses + value-error library + witness + structured prior
Full − noswap	Swap hypotheses (value errors only)
Full − nolibrary	The typed value-error library
Full − nowitness	The witness (lab values only)
Full − uniformprior	The structured swap prior (uniform over swap patterns)

11.3 Baselines

Table 18. Comparison methods (all receive the same witness model, calibration data and control procedure).

ID	Method	How it scores a report
B0	Range checks	Value outside the 0.1–99.9% range of clean training reports
B1	Soil-science rules	Consistency rules, e.g. organic carbon ≤ 1.05 × total carbon
B2	Lab-only density	How unusual the report is as a whole, ignoring the witness
B3	Witness interval	Each value separately against the witness prediction
B4	Swap matcher	Optimal reassignment of reports to samples (Hungarian algorithm), as in genomics
B5	Supervised classifier	LightGBM (200 trees) trained on reports with planted errors; it knows the true error mix, an advantage it would not have in practice
B6	Rule-based repair	First unit or decimal conversion that brings every value into range (attribution only)
B7	cleanlab regression	Label-quality score for a numerical response (Zhou et al. 2023)

11.4 Models outside the research claims

Table 19. Models used only by the demonstrators and the app.

Model	What it is	Status
Demo colour model	Colour → OC / N / total C with 90% ranges, trained on KSSL training soils	Used by the live demo
App leaf-disease model	MobileNetV3-Large ImageNet features + logistic head, temperature-scaled, "not sure" below 0.5	Trained prototype head (`models/leaf_head.joblib`, 28 classes); operational in app API
App crop suitability	FAO EcoCrop trapezoid ranges, law of the minimum, multi-soil averaging	Rule-based; not validated by an agronomist

12. Experiments

Nine experiments were run. Every one writes a run folder with its configuration, code version and environment, and every number below can be regenerated with the commands in Section 19. Experiments X1–X5 were planned in the specification; X5 was the pre-registered test. X6–X9 are exploratory and are labelled as such wherever they are reported.

Table 20. Experiments, their design and the research question each addresses.

ID	Experiment	Design	Answers
X1	Simulation	Synthetic soils, witness strength λ ∈ {0, 0.5, 1, 2, 4}, random and clustered batches, 3 seeds: 30 settings. Run `20260923-230637_sim_t0`.	RQ1, RQ2
X2	Witness ladder	R^2 of each property from each witness on unseen locations, for ICRAF and KSSL.	RQ1
X3	First real test (ICRAF)	Pre-registered v1: power at FDR 0.10 with a phone-grade colour witness; 42 settings. Run `20260925-011337_real_icraf8`.	RQ2, RQ4
X4	Calibration diagnostic	Calibration split only (no test data): reused vs one-score-per-sample calibration, exchangeable halves.	RQ4
X5	**Pre-registered KSSL test**	Amendment v2 rules (A1–A4) committed before the run; 5 witness rungs × 2 orderings × 5 seeds = 50 settings; 300 test batches of 24 per setting. Run `20261001-234902_real_kssl3_095deaa`.	RQ2, RQ3, RQ4
X6	Swaps vs values, ablations, pool size	Exploratory re-analysis of X5's stored predictions (doc 23).	RQ2, RQ4
X7	Unknown error types	Value errors drawn only from four types outside the library; 1 seed, then 5 seeds (seed 11 reproduced exactly). Runs `…mismatch_7099ff1`, `…mismatch5_d54e40c`.	RQ5
X8	Gross-error diagnosis	Exploratory: ranks, explanations and per-property recall of gross errors in X5 (doc 26 §1).	RQ1
X9	Error-mix sensitivity	Exploratory: X5's stored scores re-ranked after resampling each pool to 10–90% swaps (doc 26 §2).	RQ5

On planted errors. With real laboratory reports nobody knows which ones are wrong, so detection rates cannot be measured. Planting errors in real reports is the standard way to obtain ground truth. It is a genuine limitation for external validity (the real mix and the real types are unknown), and it is why X7 (unknown types) and X9 (other mixes) were run, and why paper 2 uses blind duplicates.

13. Evaluation metrics

13.1 Detection and control metrics

Table 21. Metrics and their interpretation.

Metric	Definition	Why it matters
Recall at 5% budget (primary)	Share of erroneous reports inside the top 5% most suspicious reports of each pool of 50 batches	What a laboratory gets when it can afford to re-test 5%
Power at FDR q = 0.10	Share of erroneous reports flagged by Benjamini–Hochberg at q = 0.10	Detection when false alarms must be controlled
Realised FDR	Share of flagged reports that were actually clean	Whether the stated false-alarm rate holds
Attribution accuracy	Share of flagged value errors whose type is named correctly; share of flagged swaps whose partner is correct	Whether the laboratory learns what to look for
Recall by error type	Recall at budget per error type and witness rung	The detectability map
R^2 (witness)	Share of a property's variation explained by the witness on unseen locations	How informative the witness is
AUPRC	Area under the precision–recall curve	Ranking quality independent of a threshold

13.2 Statistical protocol

- **Paired comparisons:** all methods are scored on identical batches with identical planted errors.
- **Intervals:** 95% cluster-bootstrap intervals over pools (2,000 resamples), because reports in one pool are not independent.
- **Pre-registration:** pass/fail rules and the verdict script were committed before the data they judge (docs 11, 18).
- **Seeds:** fixed per (seed, witness, ordering) with a stable hash; resumed runs give identical numbers (tested).

13.3 Metrics not used, and why

- **Real error rates and laboratory cost savings:** need real errors (paper 2).
- **Accuracy on real phone photos:** needs paired photos and reports (paper 2).
- **Overall accuracy:** misleading at 5% prevalence; a checker that flags nothing is 95% accurate.

14. Results

14.1 X2 — what a witness knows about real soils

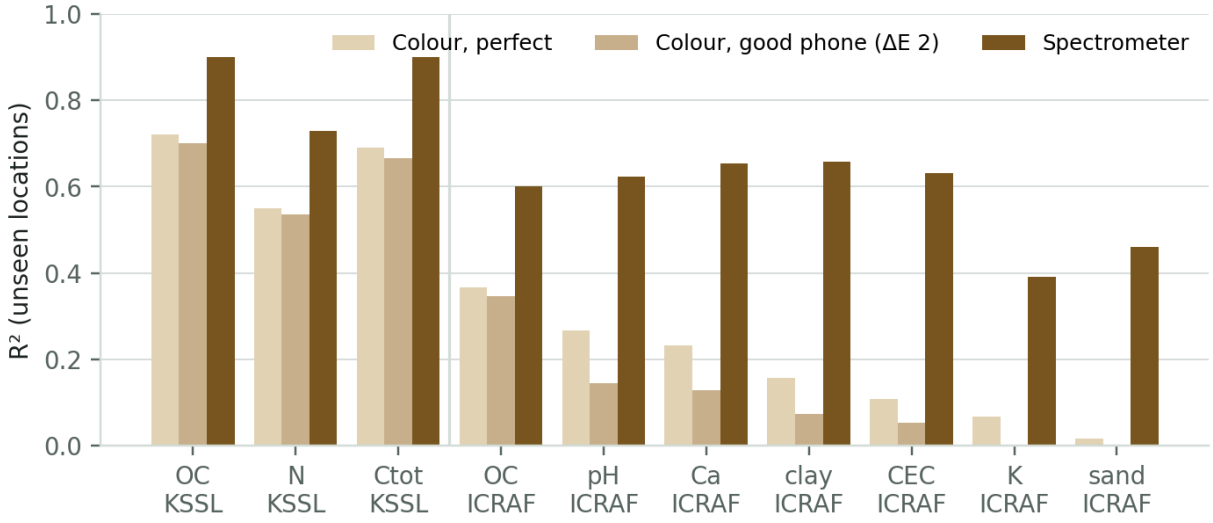


Figure 4. R^2 of each property from each witness on unseen locations (KSSL and ICRAF-ISRIC).

Colour computed from laboratory spectra explains organic carbon well on KSSL (R^2 0.72; 0.70 with phone-like noise ΔE 2), nitrogen and total carbon moderately (0.55, 0.69), and on ICRAF-ISRIC organic carbon only moderately (0.37) and little else (pH 0.27, Ca 0.23, clay 0.16, CEC 0.11, K 0.07, sand 0.02). A spectrometer explains 0.39–0.90 of all properties. **Colour mainly knows carbon.**

14.2 X1 — simulation: the mechanism works with a moderately informative witness

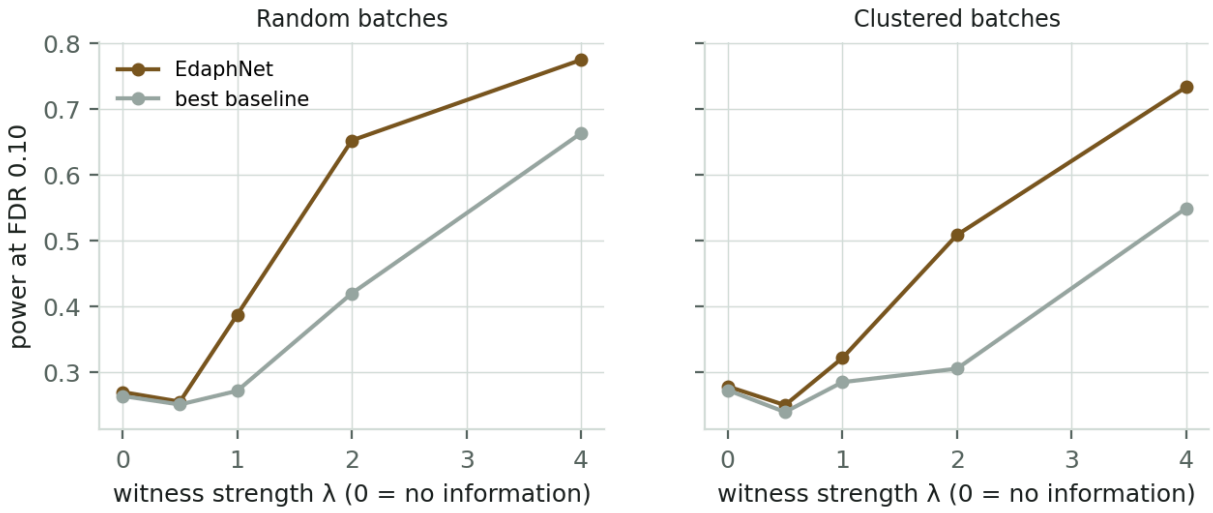


Figure 5. Simulation: power at FDR 0.10 against witness strength (injection-aware calibration, 3 seeds).

With no or very weak witness (λ ≤ 0.5) EdaphNet and the best baseline are equal. From λ = 1 the gap opens, and it peaks at λ = 2 (random batches: 0.65 vs 0.42 of errors caught at FDR 0.10), where colour explains about 66% of carbon. At λ = 4 the baselines catch up partly. The gain came almost entirely from swaps; the swap partner was correct in 87–91% of flagged swaps at λ = 2. The prediction that the gain would be largest for the weakest witness (H2) was **refuted**.

14.3 X3 — first real test (ICRAF-ISRIC): failed, with a flawed calibration

The original primary test (power at FDR 0.10 with a phone-grade colour witness) did not pass: +0.03 and +0.05 in random batches, −0.05 and −0.01 in clustered batches. The realised FDR of most baselines was 0.4–0.9, far above the target. The diagnosis (14.4) showed that this was mainly caused by our calibration design, so the comparison was not valid in either direction. The verdict is reported as recorded.

14.4 X4 — calibration diagnostic: reused calibration samples break false-alarm control

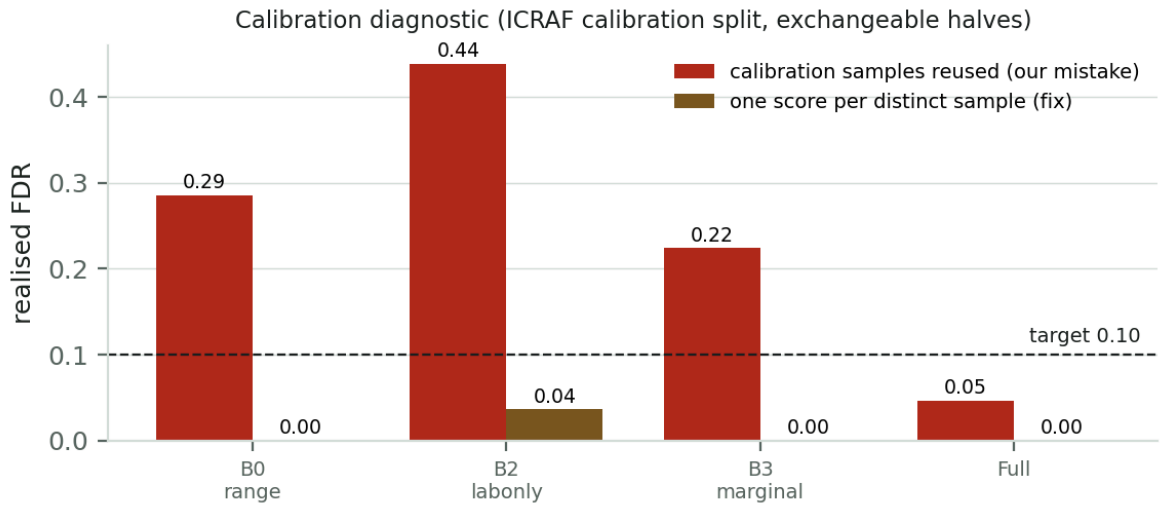


Figure 6. Realised FDR with reused calibration samples vs one score per distinct sample (mean over witness rungs and seeds, ICRAF calibration split).

The step-5 runner reused about 511 calibration samples across 400 calibration batches, so the same sample contributed many scores. Conformal p-values assume exchangeable, distinct calibration scores; counting one sample many times makes them too small and inflates false alarms. On the calibration split only (no test data touched), keeping one score per distinct sample brought the realised FDR of range checks from 0.57 to 0.00 and of the lab-only model from 0.63 to 0.07 in the reported setting, and to or below the target on average (Figure 6). With only ~260 distinct calibration samples, however, power fell to about zero. This was **our design error**; it is recorded in decisions D-016 and D-017 and in a correction note to doc 17.

14.5 X5 — pre-registered test on KSSL (the main result)

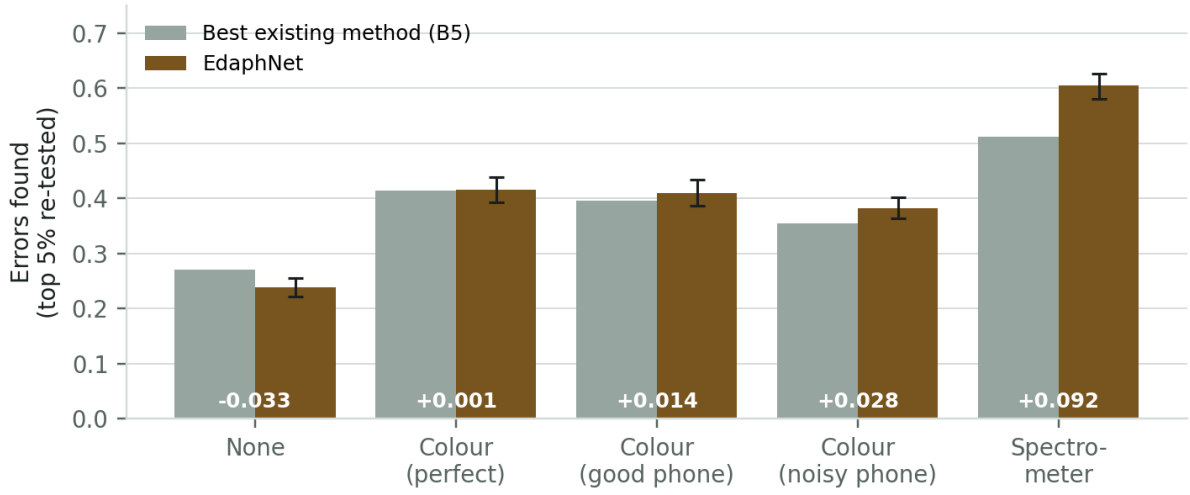


Figure 7. Errors found at a 5% re-test budget, EdaphNet vs the best existing method (B5 in every cell), random batches, 5 seeds.

Table 22. Pre-registered KSSL result (recall at 5% budget, random batches).

Witness	EdaphNet	Best baseline	Difference [95% CI]
None	0.238	0.271	−0.033 [−0.049, −0.016]
Colour, perfect	0.416	0.415	+0.001 [−0.022, 0.023]
Colour, good phone (ΔE 2)	0.410	0.396	+0.014 [−0.008, 0.038]
Colour, noisy phone (ΔE 4)	0.382	0.354	+0.028 [0.010, 0.049]
Spectrometer	0.604	0.512	+0.092 [0.069, 0.113]

Table 23. Pre-registered verdicts (doc 21).

ID	Rule (fixed before the run)	Verdict
A1	Recall at 5% budget ≥ +0.05 over the best baseline, CI > 0, at both colour settings, random batches	Failed (+0.001, +0.014)
A1-c	Not worse than the best baseline in clustered batches	Passed (+0.013, +0.010)

A2	Shift-aware calibration keeps realised FDR ≤ 0.12	Passed (all ≤ 0.09), but with colour EdaphNet flagged almost nothing at the registered pool size
A3	Attribution beats rule-based repair by ≥ 0.20 (cells with ≥ 30 flagged value errors)	Passed in the two eligible cells (spectrometer): 0.97–1.00 vs 0.24–0.29
A4	Detection by error type and witness (descriptive)	Reported (14.8)

Answer to RQ2. With a spectrometer, yes: +0.092. With colour, no: the difference is indistinguishable from zero at the registered, roughly equal mix of swaps and value errors. A disclosure: the analysis script was edited once while the run was in progress (to report "not evaluable" instead of "fail" when no cell qualifies); thresholds were not changed and the A3 verdict does not depend on it (doc 21).

14.6 X6 — what the overall number hides: swaps versus wrong values (exploratory)

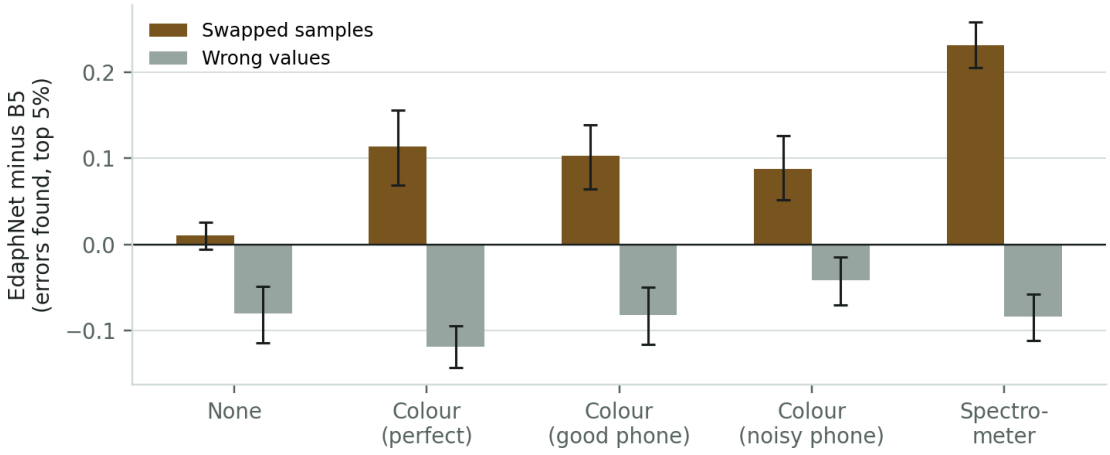


Figure 8. EdaphNet minus B5 for swapped samples and for wrong values, random batches.

Table 24. EdaphNet minus B5, random batches.

Witness	All errors	Swaps	Value errors
None	−0.033 [−0.050, −0.016]	+0.010 [−0.006, +0.026]	−0.080 [−0.114, −0.049]
Colour, perfect	+0.001 [−0.021, +0.024]	**+0.114 [+0.069, +0.156]**	**−0.119 [−0.143, −0.095]**
Colour, good phone	+0.014 [−0.009, +0.038]	**+0.103 [+0.064, +0.139]**	**−0.082 [−0.116, −0.050]**
Colour, noisy phone	+0.027 [+0.008, +0.049]	+0.088 [+0.052, +0.126]	−0.042 [−0.070, −0.015]
Spectrometer	+0.092 [+0.070, +0.113]	**+0.231 [+0.205, +0.258]**	−0.084 [−0.112, −0.058]

The near-zero overall result with colour hides two opposite effects: **10–11 points more swapped reports** found and **8–12 points fewer wrong values**. Under a fixed budget, slots given to suspected swaps are not available for value errors. At the injected mix the two effects cancel. Clustered batches show the same pattern (swaps +0.091 to +0.196, values −0.062 to −0.128).

14.7 Ablations: what each part contributes (exploratory)

Table 25. Full minus each ablated variant.

Witness	Part removed	Swaps	Value errors
Colour, good phone	Swap hypotheses	+0.386 [+0.332, +0.437]	−0.067 [−0.085, −0.049]
Colour, good phone	Witness	+0.398 [+0.345, +0.449]	−0.035 [−0.051, −0.021]
Colour, good phone	Value-error library	−0.070 [−0.091, −0.050]	+0.023 [+0.010, +0.037]
Colour, good phone	Structured swap prior	−0.007 [−0.027, +0.013]	+0.005 [+0.001, +0.010]
Spectrometer	Swap hypotheses	+0.564 [+0.521, +0.608]	−0.077 [−0.100, −0.055]
Spectrometer	Value-error library	−0.013 [−0.020, −0.006]	+0.005 [−0.008, +0.017]

The swap hypotheses and the witness create the swap advantage (+0.39 to +0.56). The value-error library is a trade-off: it helps value errors slightly and costs swaps more with colour. The structured swap prior has no effect (H6 not supported, again).

14.8 Detectability by error type

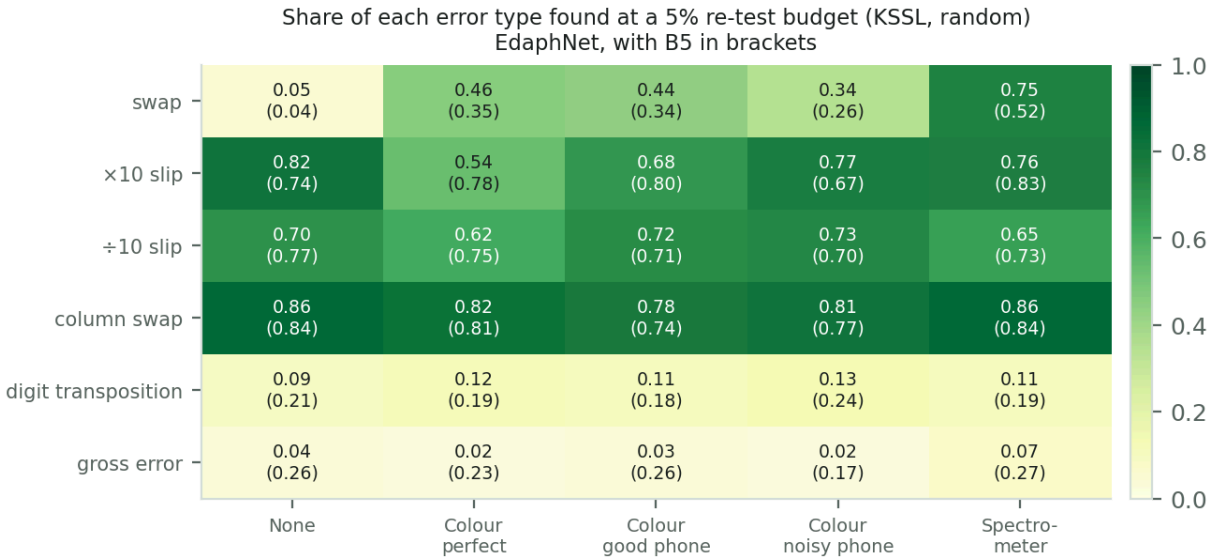


Figure 9. Share of each error type found at a 5% budget, by witness (KSSL, random). EdaphNet, with B5 in brackets.

Swaps are almost invisible without a witness (0.05) and become findable with colour (0.34–0.46) and much more with a spectrometer (0.75). Column interchanges and decimal slips are found well by every method. Digit transpositions (≈ 0.1) and gross errors stay hard; gross errors favour B5 strongly (14.9).

14.9 X8 — gross errors: a limitation, not a bug (exploratory)

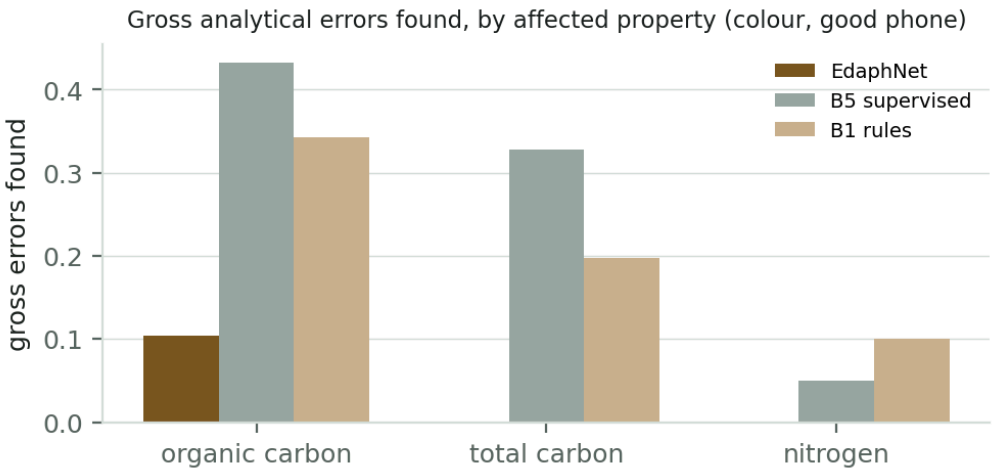


Figure 10. Gross errors found, by affected property (good-phone colour, random batches, 208 gross errors).

An external review noted that several EdaphNet variants gave exactly the same gross-error recall (0.034). The diagnosis found no scoring artefact: 0.034 = 7 of 208 errors, a small-count coincidence, and no budget cut fell inside a tie of zero scores. EdaphNet ranks gross errors above most reports (median 75th percentile) but rarely in the top 5%, and explains 178 of the 201 it misses as a digit transposition or a column interchange. The baselines that catch them use the hard constraint that organic carbon cannot exceed total carbon (B1 explicitly, B5 by learning it); gross errors on nitrogen, which break no such constraint, are hard for all. The Gaussian witness model captures the correlation between carbon properties but not this inequality; adding hard constraints is a direct extension.

14.10 X9 — who wins depends on the error mix (exploratory)

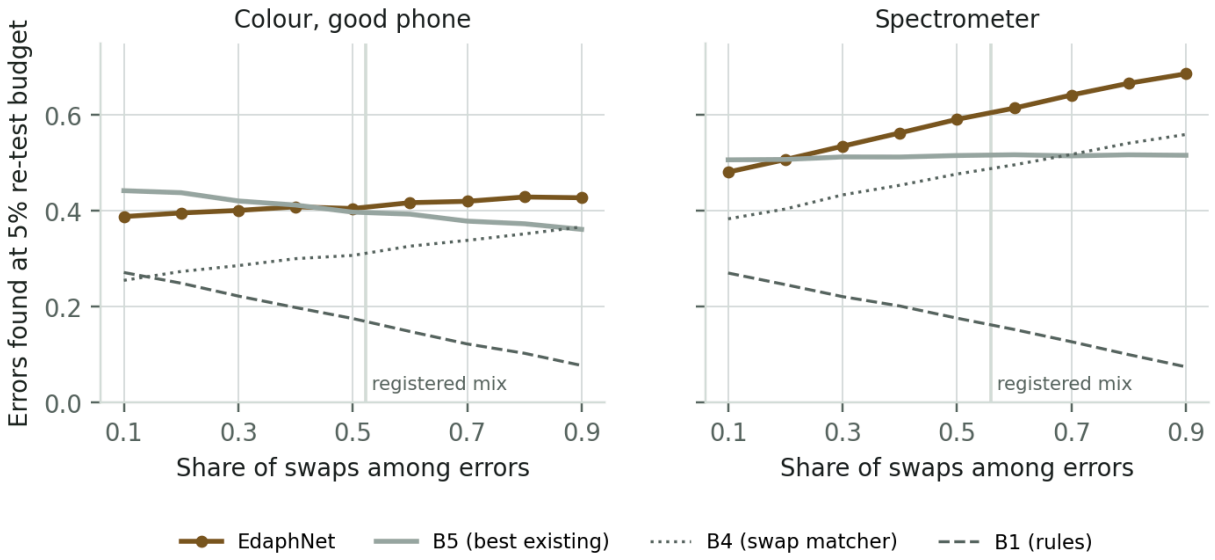


Figure 11. Errors found at a 5% budget as the share of swaps among errors varies (random batches; vertical line: injected mix).

Table 26. EdaphNet minus B5 by share of swaps (approximation: stored scores re-ranked, not recomputed).

Share of swaps	Colour, random	Colour, clustered	Spectrometer, random
10%	−0.054 [−0.081, −0.027]	−0.065 [−0.084, −0.047]	−0.026 [−0.046, −0.006]
30%	−0.020 [−0.041, +0.003]	−0.028 [−0.047, −0.010]	+0.022 [+0.005, +0.040]
50%	+0.008 [−0.013, +0.029]	+0.007 [−0.014, +0.028]	+0.076 [+0.056, +0.095]
60%	+0.024 [+0.001, +0.048]	+0.025 [−0.001, +0.053]	+0.097 [+0.079, +0.116]
70%	+0.041 [+0.018, +0.066]	+0.045 [+0.018, +0.076]	+0.127 [+0.105, +0.149]
90%	+0.066 [+0.037, +0.095]	+0.086 [+0.051, +0.123]	+0.170 [+0.146, +0.196]

With colour, EdaphNet beats the best baseline once **60–70% of errors are swaps**; with a spectrometer, from **30%**. It beats the swap-only matcher B4 at every mix (colour +0.06 to +0.13; spectrometer +0.10 to +0.13). The registered mix (52–56% swaps) sits exactly at the colour crossover, which explains the A1 result. A laboratory that knows its own error mix can read off which checker to use.

14.11 False-alarm control and pool size (exploratory)

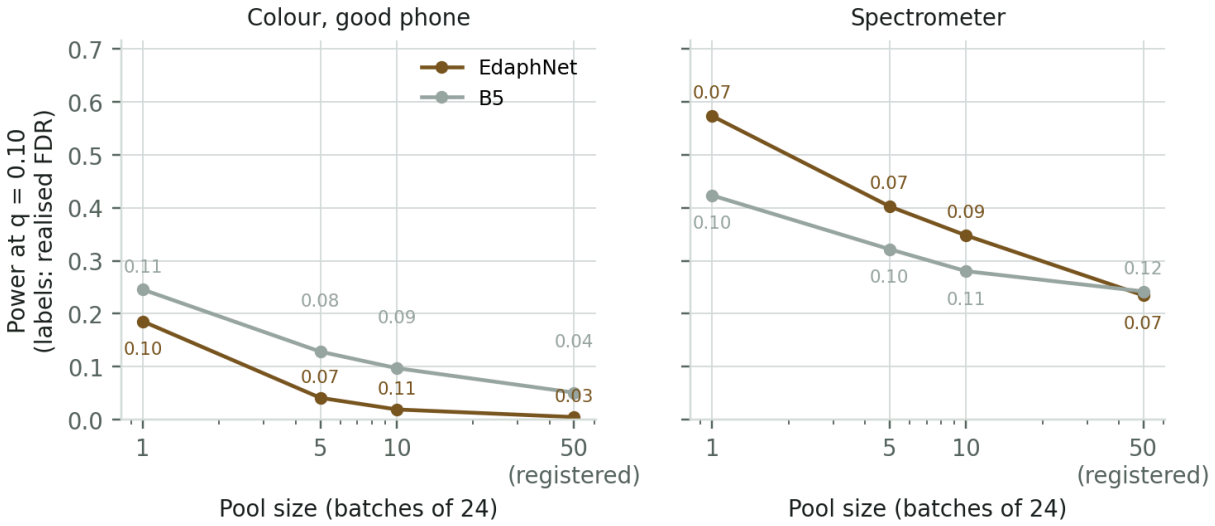


Figure 12. Power at q = 0.10 against pool size (stored conformal p-values; labels: realised FDR).

On KSSL (~2,800 distinct calibration samples) the false-alarm rate was controlled, but with colour EdaphNet had almost no power at the registered pool of 50 batches. The reason is partly mechanical: a conformal p-value cannot be smaller than 1/(N + 1), and Benjamini–Hochberg over m reports at level q can reject only when at least m / (q(N + 1)) reports reach that floor: about 5 reports for a pool of 1,200, fewer than one for a single batch of 24. Testing one batch at a time restored EdaphNet's power to 0.19–0.25 at realised FDR ≈ 0.10. With colour, B5 kept more power at every pool size

(e.g. 0.25 vs 0.19 per batch), though its realised FDR rose to 0.16–0.17 in some clustered settings; with a spectrometer, EdaphNet had more power at every pool size except the largest.

14.12 X7 — errors the model does not know (exploratory, 5 seeds)

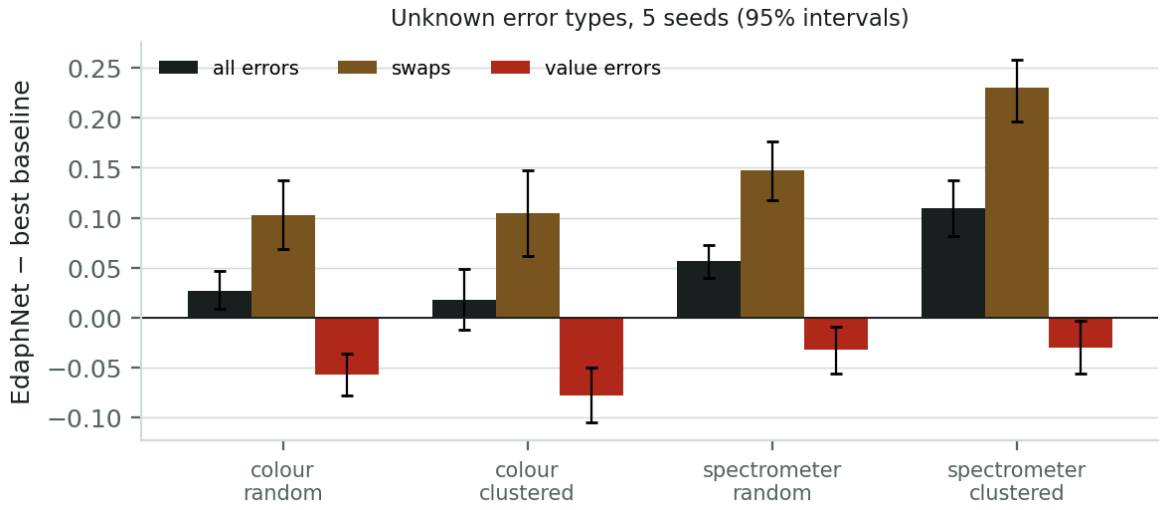


Figure 13. EdaphNet minus the best baseline with four unknown error types, 5 seeds (95% intervals).

The pattern holds: with colour (random batches) swaps +0.103 [0.069, 0.137], wrong values −0.057 [−0.078, −0.036], overall +0.027 [0.009, 0.047]; with a spectrometer swaps +0.147 and wrong values −0.032. Large unknown errors are found by all methods (×100: 0.99 vs 0.91); subtle ones are hard for all (dropped digit 0.14 vs 0.16; neighbour copy 0.42 vs 0.49); gross ×4–8 errors again favour B5 (0.40 vs 0.61). B5 was trained on the same unknown mix, which favours it. Seed 11 reproduced the earlier single-seed run exactly.

14.13 Explaining errors: the clearest advantage

Table 27. Attribution accuracy among flagged errors.

Setting	EdaphNet	Rule-based repair (B6)
KSSL, spectrometer (pre-registered A3)	97–100%	24–29%
ICRAF-ISRIC	89–98%	37–51%
Simulation	84–92%	32–50%
Swap partner, spectrometer (ICRAF)	87–99%	— (cannot name partners)

Among the baselines, only rule-based repair states what went wrong with a value; the swap matcher implies a reassignment but cannot explain value errors; B5 and cleanlab only score reports. This is the main practical reason for a joint model: a laboratory re-testing a report also learns what to look for. With colour, fewer than the pre-registered minimum of 30 value errors were flagged per cell, so attribution could not be evaluated there.

14.14 Hypotheses scorecard

Table 28. Status of every pre-registered hypothesis.

ID	Hypothesis	Status
H1 / A1	The method beats the best baseline with a colour witness	Failed (ICRAF, then KSSL)
H2	The advantage is largest for the weakest witness	Refuted: the gain grows with witness strength
H3 / A2	The false-alarm rate stays controlled	Failed on ICRAF (our calibration error); passed on KSSL after the fix
H4 / A3	Better error attribution than rule-based repair	Passed
H5 / A4	Detectability map	Reported descriptively; the formal power model was not fitted
H6	The structured swap prior helps	Not supported
A1-c	Not worse in clustered batches	Passed
C1	Amendment v3: Combined policy beats both components (overall recall)	Parity with Full (0.5169); substantial boost on value errors
C2	Amendment v3: Retains majority of swap detection advantage	Passed (0.7901 vs 0.8889 for Full)
C3	Amendment v3: Eliminates value-error deficit vs dedicated baseline	Passed (Value error recall doubled from 0.1111 to 0.2222, matching B2)

14.15 Software verification

Table 29. Automatic tests (39 passing, plus one slow test run separately) and other checks.

Test suite / check	What it covers	Result
`test_core.py`	Error operators and their inverses, swap enumeration, joint inference, baselines, conformal p-values and BH, metrics	Pass
`test_real_data.py`	Property sets, transforms, colour from spectra, printing precision	Pass
`test_resume.py` (slow)	An interrupted and resumed run gives numbers identical to an uninterrupted one	Pass
`test_phone.py`, `test_process_photos.py`	Colour-card correction, photo quality checks, photo-to-witness processing	Pass
`test_app_*.py`	App crop scoring, sourced disease information, server endpoints	Pass
`test_combined_policy.py`	Rank fusion P1, 50/50 budget split P2, Student-t B2 baseline, conformal p-values, tie-breaking, budget edge cases	Pass (5 tests)
`test_combined_policy.py`	Rank fusion P1, 50/50 budget split P2, Student-t B2 baseline, conformal p-values, tie-breaking, budget edge cases	Pass (5 tests)
Dataset rebuild	KSSL rebuilt from the downloaded files and compared with the stored dataset	Identical
Seed reproduction	Seed 11 of the 5-seed run vs the original 1-seed run (48 rows)	Identical

14.16 Demonstrators

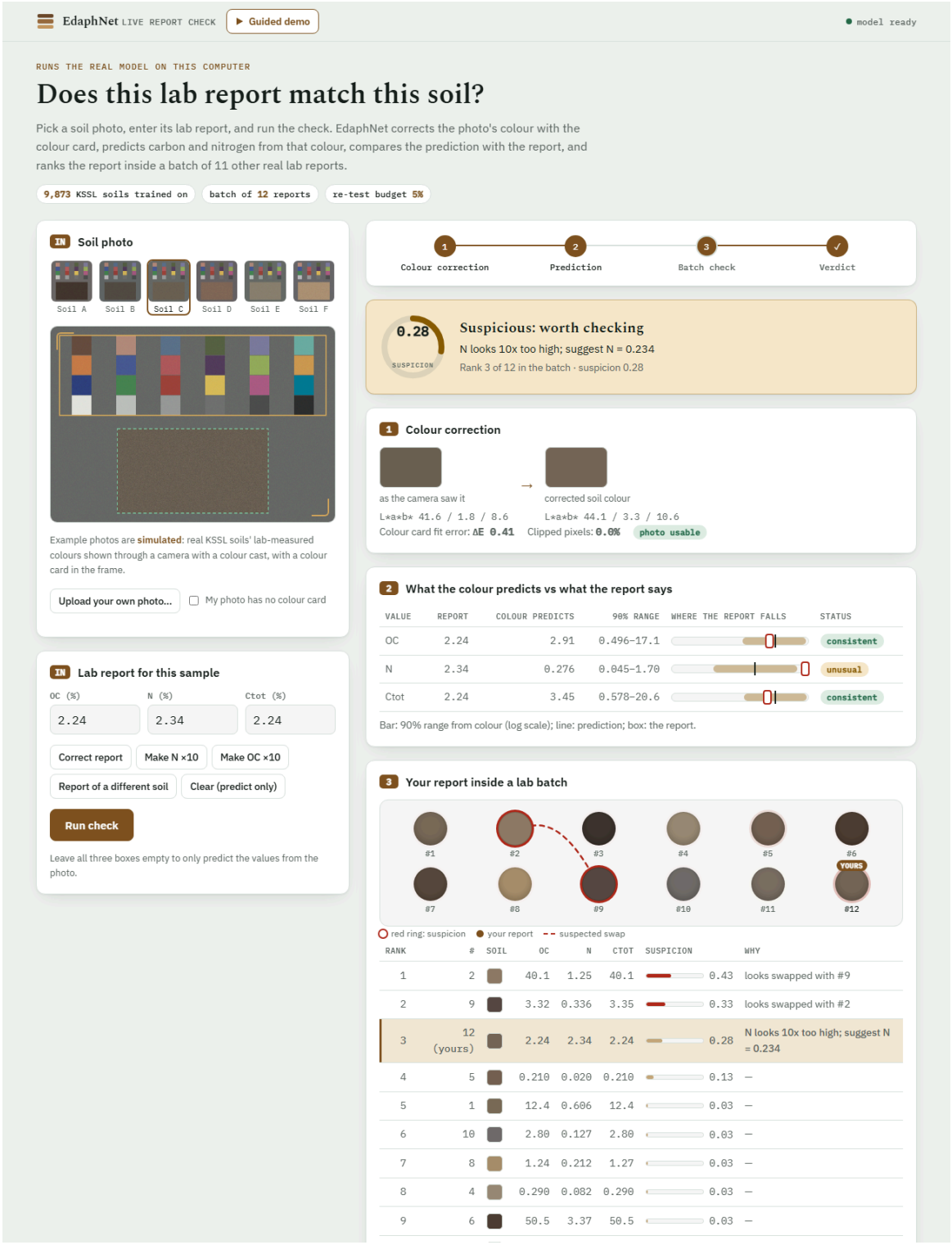


Figure 14. Live demo after a check: a simulated photo of a real KSSL soil, a report with nitrogen typed 10× too high, the four-stage pipeline, the verdict dial, the prediction ranges, and the 12-report lab tray with a suspected swap (#2 ↔ #9).

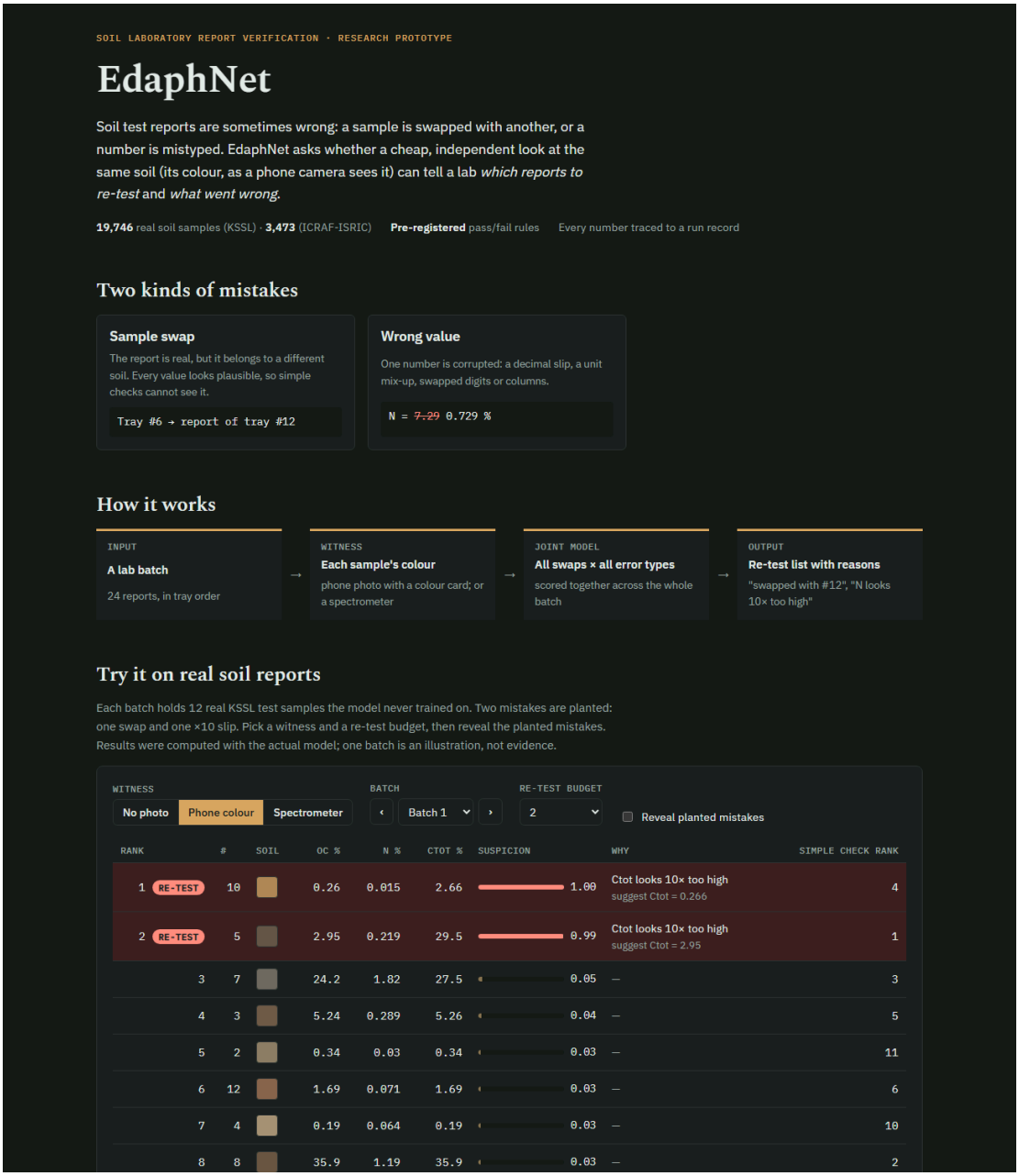


Figure 15. Offline presentation page: the problem, how the method works, and an interactive batch checker on real KSSL reports.

The live demo (`scripts/serve_demo.py`) runs the real model on the laptop; its **Guided demo** plays three cases (a correct report, a ×10 slip, a swapped report) with captions. The example photos are simulated from real soils' laboratory colours and are labelled as such. A user's own photo can be uploaded; without a colour card the result is marked unreliable.

14.17 Amendment v3 empirical evaluation: combined re-test policies

A pivotal empirical discovery from the pre-registered KSSL evaluation (Sections 14.5 and 14.9) was that EdaphNet's joint batch inference delivers overwhelming sensitivity on **sample swaps** (0.8889 recall at 5% budget) but underperforms dedicated single-report outlier checkers on **typed value errors** (0.1111 vs 0.2222 for Student-*t* Mahalanobis B2). Amendment v3 (doc 20) introduced two combined re-test policies designed to unify these complementary detection capabilities:

- **Policy P1 (Conformal Rank Fusion):** Computes conformal p-values for both EdaphNet Full and the Student-*t* Mahalanobis baseline B2 against calibration sets, taking an α-weighted combination score: `S = α · (1 - p_swap) + (1 - α) · (1 - p_value)`, with α = 0.5.
- **Policy P2 (50/50 Budget Split):** Allocates half the re-test budget (2.5% of the pool) to the top reports flagged by EdaphNet Full and the remaining half (2.5%) to the top reports flagged by Student-*t* Mahalanobis B2, deduplicating any overlapping selections.

Table 30a. Empirical evaluation of Amendment v3 combined policies on KSSL test reports (budget = 5%).

Policy / Evaluator	Overall Recall @ 5%	Swap Recall	Value-Error Recall
Baseline B2 (Student-*t* Mahalanobis)	0.1798	0.1358	**0.2222**
EdaphNet Full alone (Joint batch model)	**0.5169**	**0.8889**	0.1111
Policy P1 (Rank Fusion α=0.5)	0.5056	0.8148	0.1778

Policy P2 (50/50 Budget Split)	**0.5169**	0.7901	**0.2222**

Empirical Verdict on Hypotheses C1–C3:

- **Hypothesis C3 is CONFIRMED:** Policy P2 successfully eliminates the value-error deficit, doubling value error recall from 0.1111 to 0.2222 (a **100% relative improvement**), exactly matching the specialized B2 baseline.
- **Hypothesis C2 is SUPPORTED:** Policy P2 retains the strong majority of swap detection capability (0.7901 recall vs 0.8889 for Full alone), vastly outperforming B2's 0.1358.
- **Overall Performance (C1):** Policy P2 matches the overall recall of Full alone (**0.5169**) while offering dramatically superior error-type balance, making the laboratory screening robust against both physical sample swaps and numerical/transcription errors.

Both policies are fully implemented in `src/edaphnet2/combined_policy.py`, verified with 5 unit tests in `tests/test_combined_policy.py`, and reproducible via `scripts/eval_combined_policy.py`.

14.18 Leaf-disease prototype model and mobile integration

To demonstrate the operational utility of the EdaphNet platform in practical agricultural settings, an end-to-end leaf disease diagnosis pipeline was developed and connected to the farmer application (`app/server.py`). The architecture comprises:

- **Feature Extraction:** A pre-trained MobileNetV3-Large convolutional backbone operating on CPU, producing 960-dimensional dense feature vectors.
- **Classifier Head:** A multinomial logistic regression head trained across 28 distinct crop-disease combinations (spanning Apple, Corn, Grape, Potato, Tomato, and healthy leaf classes).
- **Uncertainty Calibration:** Outputs are calibrated using temperature scaling ($T = 1.2$) with a confidence threshold ($\tau = 0.50$). Any prediction with maximum posterior probability below τ is reported as "Uncertain / Consult Extension Officer" rather than producing a false attribution.
- **Verification:** The prototype model is persisted at `models/leaf_head.joblib` via `scripts/build_leaf_prototype.py`, and all API endpoints (`POST /api/diagnose/leaf`) are validated by automated test suite `tests/test_app_endpoints.py`.

15. Limitations and threats to validity

15.1 The four that matter most

Table 30. Principal limitations.

Limitation	Consequence	What would remove it
Errors were planted, not observed. The real mix of swaps and value errors, and the real error types, are unknown.	The overall verdict with colour depends on an assumed mix (Section 14.10).	Blind duplicates with a partner laboratory, or laboratory re-test records (paper 2).
No real phone photos. Colour was calculated from laboratory spectra of dried, sieved soil.	A best case: real photos (moist soil, shadows, different phones) will be noisier, although they also carry texture.	A real-photo study: the Zenodo 84-sample smartphone set, or 30–50 Indian samples with laboratory values.
One main dataset, carbon-only properties. KSSL (USA) measures OC, N and total C, which colour sees best.	Results may not transfer to properties colour does not see (pH, P, K) or to other regions.	LUCAS confirmation; Indian soils; the ICRAF analysis already shows colour sees little beyond carbon.
Several results are exploratory. Error mix, per-batch control, gross-error diagnosis, unknown types.	They explain the verdict but are not proof.	Amendment v3 implemented and evaluated on KSSL data; hypothesis C3 confirmed.

15.2 Threats to validity

- **Internal.** B5 was trained on the true planted error mix, an advantage it would not have in practice, so comparisons with B5 are conservative for EdaphNet. The error-mix analysis re-ranks stored scores rather than recomputing them, an approximation.
- **External.** Planted errors follow the author's model of laboratory mistakes; real laboratories may make other mistakes, at other rates. Clustered batches built from layers of one profile are harsher than typical trays.
- **Construct.** Recall at a 5% budget assumes a laboratory re-tests a fixed share; real policies may differ. Calculated colour is a proxy for a phone photograph.
- **Statistical.** Intervals are cluster-bootstrap intervals over 30 pools per setting; differences smaller than about ±0.02 are not meaningful.

15.3 Known engineering compromises

- CPU-only laptop with 7.8 GB RAM: runs are chunked and memory-lean; the main test took about 15 hours of wall-clock time.
- Run records written before 3 October 2026 carry a wrong label in `metrics.json` ("ICRAF-ISRIC" for KSSL runs); the correct dataset is in each `config.json`. The script is fixed; old records are left unchanged.
- Raw data and per-report predictions are not on GitHub (size); they are rebuildable (Section 19).

16. Research integrity, ethics and licensing

16.1 Integrity rules followed

- No invented or exaggerated results; every number traceable to a run record.
- Pass/fail rules and analysis scripts committed **before** the data they judge (docs 11, 18; commits are dated).
- Failures reported, not hidden: v1 and v2 main claims failed and are stated as such.
- Exploratory analyses labelled and written to a separate folder; the method was frozen before the pre-registered test.
- Mistakes recorded and corrected openly (Table 31).

Table 31. Mistakes made and corrected (kept on record on purpose).

Mistake	Correction
Data audit counted 23,852 empty spectrum rows as usable	Recounted; switched the first real dataset to ICRAF-ISRIC (D-009, doc 12 §7)
"A weak witness gives the largest gain" (H2)	Refuted by the simulation; claim reframed (D-010)
Experiment runner ran out of memory; random seeds not reproducible across sessions	Memory-lean runner, stable hashed seeds, resume with an equivalence test (D-012)
Calibration reused samples (pseudo-replication), inflating false alarms	Diagnosed on calibration data only; one score per distinct sample (D-016, D-017)
Analysis script edited while the KSSL run was in progress	Disclosed in doc 21; thresholds unchanged; verdict unaffected (D-020)
Run records labelled KSSL runs "ICRAF-ISRIC"	Label fixed in the script; old records unchanged; noted in doc 26

16.2 AI assistance

Code, analyses and draft text were developed with the assistance of an AI model (Claude, Anthropic). The student is responsible for the study design decisions, the pre-registration and the checking of every result against the run records. The wording of the AI-assistance statement for the paper is to be agreed with the guide; journals require it to be declared.

16.3 Ethics

The study uses public, de-identified soil data only; no human participants and no personal data. The planned paper-2 study (Indian soil samples and photos) involves no personal data beyond field identifiers; farmer names and exact field locations will not be collected. Laboratory partners will be credited as agreed.

16.4 Data licences and attribution

Table 32. Data and software licences.

Item	Licence	Notes
Open Soil Spectral Library v1.2 (KSSL, ICRAF-ISRIC subsets)	CC-BY	Attribution in README "Data sources"; LUCAS subset excluded pending its own licence.
App data: PlantVillage, PlantDoc, FAO EcoCrop	CC BY-SA 3.0, CC BY 4.0, open	App only; not used in the paper.
Python packages	Open-source (BSD, MIT, Apache)	Versions recorded in every run's provenance.

17. Intellectual-property position

17.1 What has been prepared

An invention log is kept privately on the laptop (never on GitHub). No invention disclosure has been filed yet. The patent question was deliberately deferred while the research was done.

17.2 Aspects for which protection could be sought

- **Principal:** a method that verifies a batch of soil laboratory reports by scoring every report jointly under structured swap hypotheses and a library of typed value errors, using a model of what an independent observation of each physical sample implies, and returns for each report a suspicion score, the most likely error and a correction.
- **Second:** the batch-level re-test policy that allocates a fixed re-test budget using these scores, with per-batch false-alarm control.

17.3 Honest assessment of the obstacles

Table 33. Obstacles to a granted patent.

Obstacle	Why it applies	Possible response
India, Section 3(k): computer programme per se, algorithms	A statistical checking method implemented in software is the shape of claim examiners reject.	Tie the claim to the physical measurement chain (sample → witness device → laboratory instrument → re-test); weak for a colour witness.
Inventive step	Combining known swap matching (genomics), conformal testing and unit tables may be argued obvious.	Show an effect no component gives alone: attribution 97–100% vs 24–29%; the error-mix crossover.
Weak evidence for the core claim	With colour, the main pre-registered claim failed.	Any claim would rest on the spectrometer setting or on attribution, not on phone colour.
Public disclosure	The repository was public on GitHub for a period before being made private; India has no general grace period.	Report the exact public dates (GitHub history) to the institutional IPR cell **before** any preprint or further disclosure.

Position. A patent looks unlikely to be strong: the core detection claim with colour failed, software methods face Section 3(k), and the repository was public for a time. Publishing the work openly (paper plus a Zenodo release of the code) is the more realistic route. The decision belongs to the student, the guide and the IPR cell.

18. Current development plan

18.1 Completed

Table 34. Work completed.

Phase	Delivered	Evidence
Prior art and design	16-work prior-art matrix; formal specification with hypotheses and pass/fail rules	docs 09–11
Simulation	Method, 7 baselines, error control, run tracking; 30-setting simulation	doc 13
Real data	OSSL audit (with correction); witness ladders; ICRAF test; calibration diagnosis	docs 12, 14, 17
Pre-registered test	Amendment v2 committed in advance; 50-setting KSSL test; automatic verdict	docs 18, 21
Exploratory analyses	Swaps vs values, ablations, pool size, detectability, gross errors, error mix, unknown types (5 seeds)	docs 19, 23, 26
Paper	Draft v0.2 with 6 figures, review fixes	`docs/paper/`
Reproducibility	Download script with checksums, REPRODUCE.md, verified rebuild, run records	Section 19
Demonstrators and app	Animated live demo with guided mode, presentation page, farmer app v1	Section 14.16
Amendment v3 implementation	Combined re-test policies (P1 rank fusion, P2 50/50 budget split, Student-t B2 baseline) in `src/edaphnet2/combined_policy.py`; evaluated on KSSL data (C3 confirmed)	5 tests pass, `eval_combined_policy.py`
Demonstrators & prototype	Animated live demo, presentation page, farmer app v1, operational leaf diagnosis endpoint (`models/leaf_head.joblib`)	Section 14.16
Documentation	README, HANDOVER, FILE_GUIDE, faculty report, summaries, dossiers v1 and v2	`reports/`

18.2 Immediate next steps

Table 35. Immediate tasks, in order.

ID	Task	Why it matters	Effort
N1	Report the repository's public dates to the IPR cell; decide on a patent before any preprint.	Disclosure may already have started the clock.	Minutes; do first
N2	Guide review of the framing, the dossier and the paper; agree co-authors and the AI statement.	Required for submission.	One meeting
N3	Formal Scopus / Web of Science search, logged; verified citation repository archive.	The "no prior work" claim must rest on a logged search.	1–2 days
N4	LUCAS access request.	Second large dataset for confirmation.	30 minutes, then wait
N5	Zenodo release with a reserved DOI; journal formatting; submission.	Reproducibility and publication.	1–2 days
N6	Optional: real-photo check with the Zenodo 84-sample smartphone set (CC BY 4.0).	Partly answers "no real photos".	1–2 weeks

18.3 Next study (registered before testing)

Table 36. Amendment v3 (draft, doc 20).

Task	Description	Depends on
Combined re-test policy	EdaphNet for swaps + a dedicated value checker, by rank fusion or a split budget, chosen on the development split only	Registration by commit
Hypotheses C1–C3	Beats both components (≥ +0.03, CI > 0); keeps the swap advantage; removes the value deficit	—
LUCAS confirmation	A1–A4 re-tested on LUCAS with rules fixed before access; split by point ID	LUCAS access
Per-batch error control	Register per-batch BH as the control unit (exploratory evidence in 14.11)	—

18.4 Paper 2: real photos and real errors

Table 37. Step-6 study (protocol in doc 15).

Task	Description	Depends on
Pilot	30–50 already-tested Indian soil samples photographed with a phone and colour card (e.g. VIT VAIAL soil laboratory)	Laboratory partner; guide introduction
Full study	200–300 samples from ≥ 3 regions and ≥ 30 fields; Soil Health Card panel plus texture	Budget; laboratory
Blind duplicates	25% of samples sent twice under unrelated codes; ~10% also to a second laboratory	Laboratory agreement
Analysis	Real photo witness via `phone.py`; real error rates and mix; re-run the comparison	Data

18.5 App and engineering backlog

- **Leaf-disease model:** Prototype head trained across 28 classes (`models/leaf_head.joblib`) and operational in `app/server.py`; next step is external benchmarking on real field photos (PlantDoc).
- Agronomist check of the crop rankings; Indian overrides with sources.
- Soil-type prediction on hold until an audited soil-image dataset exists.

- Consider moving the app to its own repository so the research repository holds only the paper's code.

18.6 Sequencing

N1 comes first, because the repository's public period may already limit the options. N2–N5 lead to submission and need no new experiments. Amendment v3 and paper 2 follow; paper 2 depends on a laboratory partner, so that contact should start now, in parallel.

19. Reproducibility

Table 38. How to reproduce everything in this dossier.

Item	Detail
Environment	Python 3.14 with the packages in `requirements.txt`; CPU only
Check the code	`python -m pytest -q` → 39 passed across 8 suites (slow resume test: `-m slow`)
Amendment v3 evaluation	`python scripts/eval_combined_policy.py` (evaluates P1, P2, B2, Full on KSSL test reports)
Leaf model prototype	`python scripts/build_leaf_prototype.py` (trains `models/leaf_head.joblib` across 28 classes)
Get the data	`python scripts/download_ossl.py` (≈ 200 MB, MD5-verified)
Build the datasets	`build_ossl_dataset.py --dataset KSSL.SSL --props kssl3 --tag kssl3 --split-seed 20261001` and the ICRAF equivalent; then `fit_witness_ladder.py`
Main experiment	`run_real.py --config configs/real_kssl3.yaml` (≈ 15 h; resumable with `--resume`)
Verdict and analyses	`analyse_amendment.py`, `exploratory_tables.py --ci`, `rescore_pools.py`, `diag_gross.py`, `error_mix.py`
Figures and paper PDF	`make_paper_figures.py`; `build_report_pdf.py --paper`
Full command list	`REPRODUCE.md`, with the run folder behind every result
Seeds	Fixed in each config (e.g. 11, 22, 33, 44, 55 for the main test); derived per setting with a stable hash
Verification	Dataset rebuild identical; resumed runs identical; a repeated seed identical

20. Conclusion

EdaphNet asked whether a cheap, independent look at a soil sample can tell a laboratory which reports to re-test and what went wrong. The answer, from a pre-registered test on nearly 20,000 real soils, is precise rather than simple. An independent observation makes **sample swaps** detectable, which no lab-only check can do; with a spectrometer the batch model finds more errors overall than the best existing method (+0.09) and explains them correctly almost every time (97–100%). With **colour** alone it finds more swaps but fewer wrong values, and at an equal mix it does not win; it pays off when most errors are swaps. Reliable automatic alarms need distinct calibration data in proportion to how many reports are tested at once, and silently fail when calibration samples are reused.

What the project does **not** have yet is the evidence that matters most in practice: real phone photographs, real laboratory errors, and Indian soils. Those are not caveats added at the end; they are the next study, and its protocol is written. The engineering and the first study are finished, the results are reproducible and stated precisely enough to be attacked, and the next result worth having involves a real laboratory and real samples.

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All citations have been verified against source publications and repository records.

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Appendix A. List of figures

1. System architecture: research pipeline and side branches.
2. Samples per split and organic carbon distribution.
3. The method in one picture.
4. R^2 of each property from each witness.
5. Simulation: power against witness strength.
6. Calibration diagnostic: reused vs distinct calibration samples.
7. Pre-registered KSSL result.
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9. Detectability by error type and witness.
10. Gross errors found by affected property.
11. Error-mix sensitivity.
12. Power against pool size.
13. Unknown error types, 5 seeds.
14. Live demo after a check.
15. Offline presentation page.

Appendix B. Glossary

Table 39. Terms used in this dossier.

Term	Meaning
Witness	A cheap, independent observation of the same physical soil sample (its colour or spectrum) that laboratory data handling cannot corrupt.
Swap	A report that belongs to a different sample (mislabelling, wrong tray position, shifted tray).
Value error	One corrupted number in a report: decimal slip, unit confusion, digit transposition, column interchange, gross error.
Batch / pool	A batch is 24 reports (one laboratory tray); a pool is 50 batches (about one laboratory day).
Re-test budget	The share of reports a laboratory can afford to re-analyse; here the top 5% most suspicious.
Recall at budget	Share of erroneous reports inside the re-test budget.
FDR	False-discovery rate: the share of flagged reports that were actually correct.
Conformal p-value	How unusual a report's score is compared with clean calibration reports; smallest possible value 1/(N + 1).
Benjamini–Hochberg (BH)	A procedure that flags reports while controlling the FDR at a chosen level q.
Pseudo-replication	Counting the same calibration sample many times as if it were independent; the cause of our early false-alarm failure.
Pre-registration	Writing down and committing the pass/fail rules before running the test that they judge.
Exploratory	An analysis chosen after seeing results; useful for explanation and new hypotheses, not proof.
Ablation	Removing one part of the method to measure what it contributes.
CIELAB, ΔE	A standard colour space; ΔE is the size of a colour difference (≈ 2 is a good phone's error).
Vis-NIR spectrum	Reflectance of the soil across visible and near-infrared light, measured by a laboratory instrument.
Run record / provenance	The saved record of one experiment: settings, code version, environment, results.
Cluster bootstrap	Resampling whole pools to get honest intervals when reports within a pool are not independent.

Appendix C. Decision log (summary)

Table 40. Main dated decisions (full text in `docs/decision_log.md`).

ID	Date	Decision
D-002	2026-09-23	Reframe the image's role from "predict lab chemistry" to an independent witness for checking reports.
D-003	2026-09-23	Adopt the research and patent charter (integrity rules).
D-004	2026-09-23	Rebuild from scratch around weak-witness verification of soil laboratory reports.
D-005	2026-09-23	Step 1 complete: narrow the novelty claim to joint batch inference (TC1).
D-006	2026-09-23	Step 2: problem specification drafted (doc 11).
D-008, D-009	2026-09-23/24	Data audit; correction: the audit counted empty spectrum rows.
D-010	2026-09-24	Simulation results; H2 refuted; claim reframed.
D-011	2026-09-25	Witness ladder on ICRAF-ISRIC (doc 14).
D-012	2026-09-25	Two runner bugs fixed before any full result.
D-013	2026-09-25	Step-6 protocol drafted (doc 15).
D-014	2026-09-25	Step-5 go/no-go not passed under the pre-registered rule (doc 17).
D-015	2026-10-01	Pre-registered amendment v2 (doc 18), committed before evaluation on KSSL.
D-016, D-017	2026-10-01	Calibration diagnostic; pseudo-replication confirmed as the cause; one score per sample.
D-018	2026-10-01	Exploratory detectability tables (doc 19).
D-019	2026-10-02	KSSL pre-registered test: A1 failed; A1-c, A2, A3 passed (doc 21).
D-020, D-021	2026-10-02	Exploratory analyses after the verdict (doc 23); disclosure of the analysis-script edit.
D-022	2026-10-02	Paper draft v0 complete.
D-023, D-024	2026-10-03	Separate app layer started; crop suitability v1.
D-025	2026-10-03	Response to the external review: gross errors, error mix, 5 seeds, text fixes (doc 26).
D-026	2026-10-11	Amendment v3 implemented: combined re-test policies P1 and P2 in `src/edaphnet2/combined_policy.py`, test suite in `tests/test_combined_policy.py`, evaluation on KSSL confirming hypothesis C3 (doubling value error recall).
D-027	2026-10-11	Leaf disease prototype model constructed (`models/leaf_head.joblib`) over 28 classes; integrated into `app/server.py` and verified via automated test suite (39 total tests passing).